

The story of

Push & Pull

Written By

Ahmed Moawad



Part 1

Introduction

Once upon a time, There is a newspapers seller called waheed. He has a newspaper shop and wake up every morning to sell fresh newspapers.



4to90

Asynchronous vs Synchronous

Dear <Place holder>,

I'm your Martian brother and I came to
save you from them.

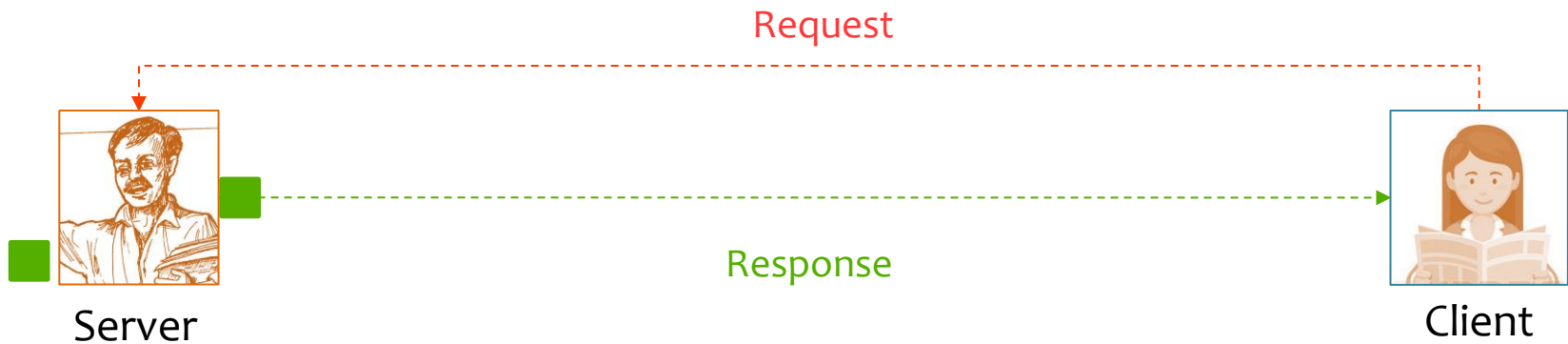
Sincerely,
1100293



Part 2

Pull

Every morning, Amina go to waheed's newspaper market and buy a newspapers then go to her work.



AJAX

AJAX stands for Asynchronous JavaScript And XML

With **AJAX** you can do the following



Update a web page without reloading it.



Request and **receive data** from a server after the page has loaded.



Send data to a server in the background.



How does it work ?



Client

Get My Name



Server



How does it work ?



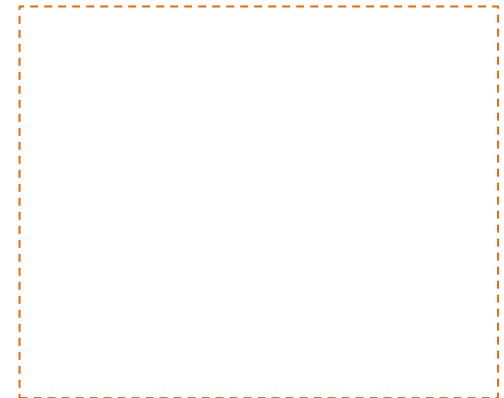
Client

Get My Name

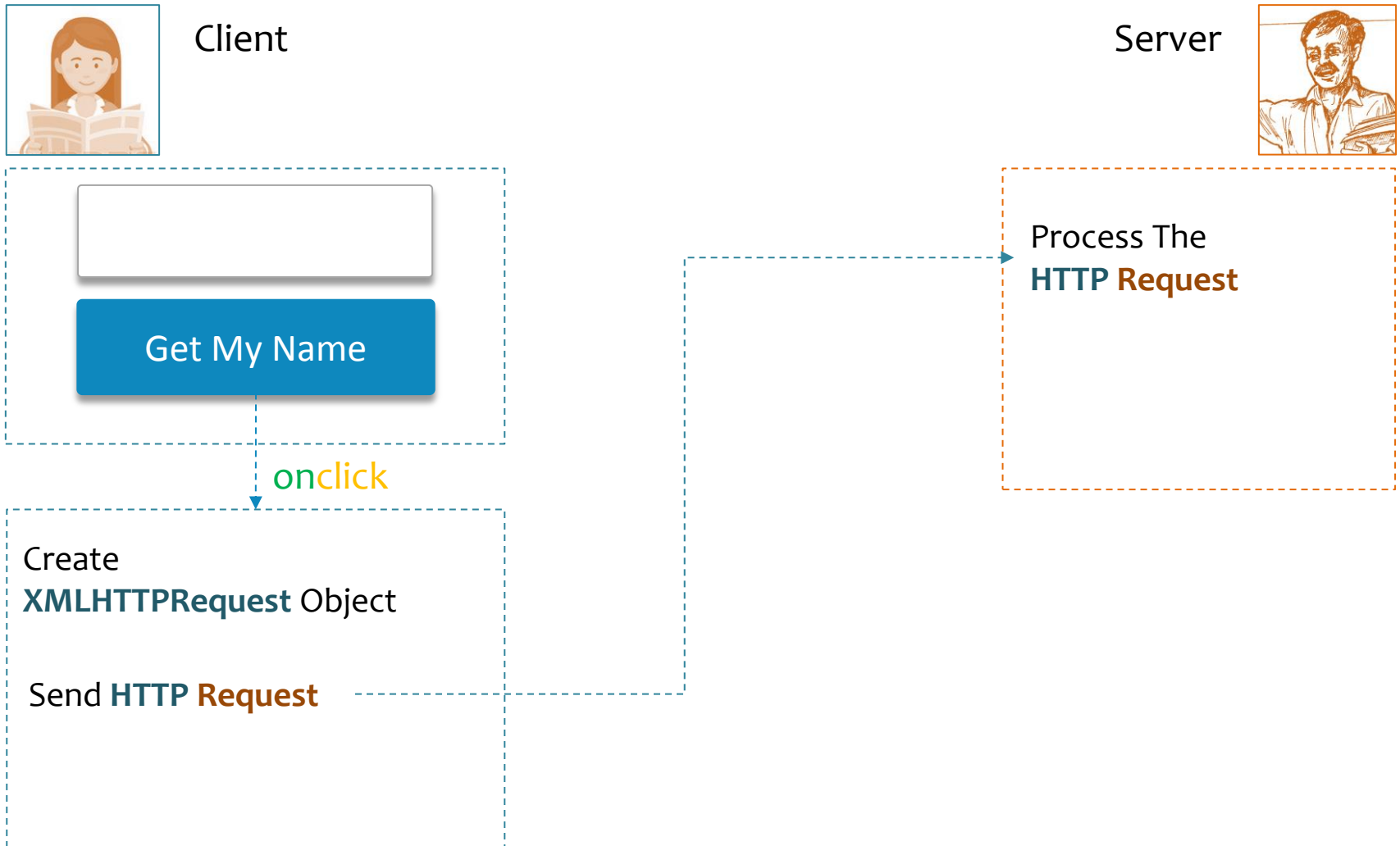
onclick

Create
XMLHttpRequest Object

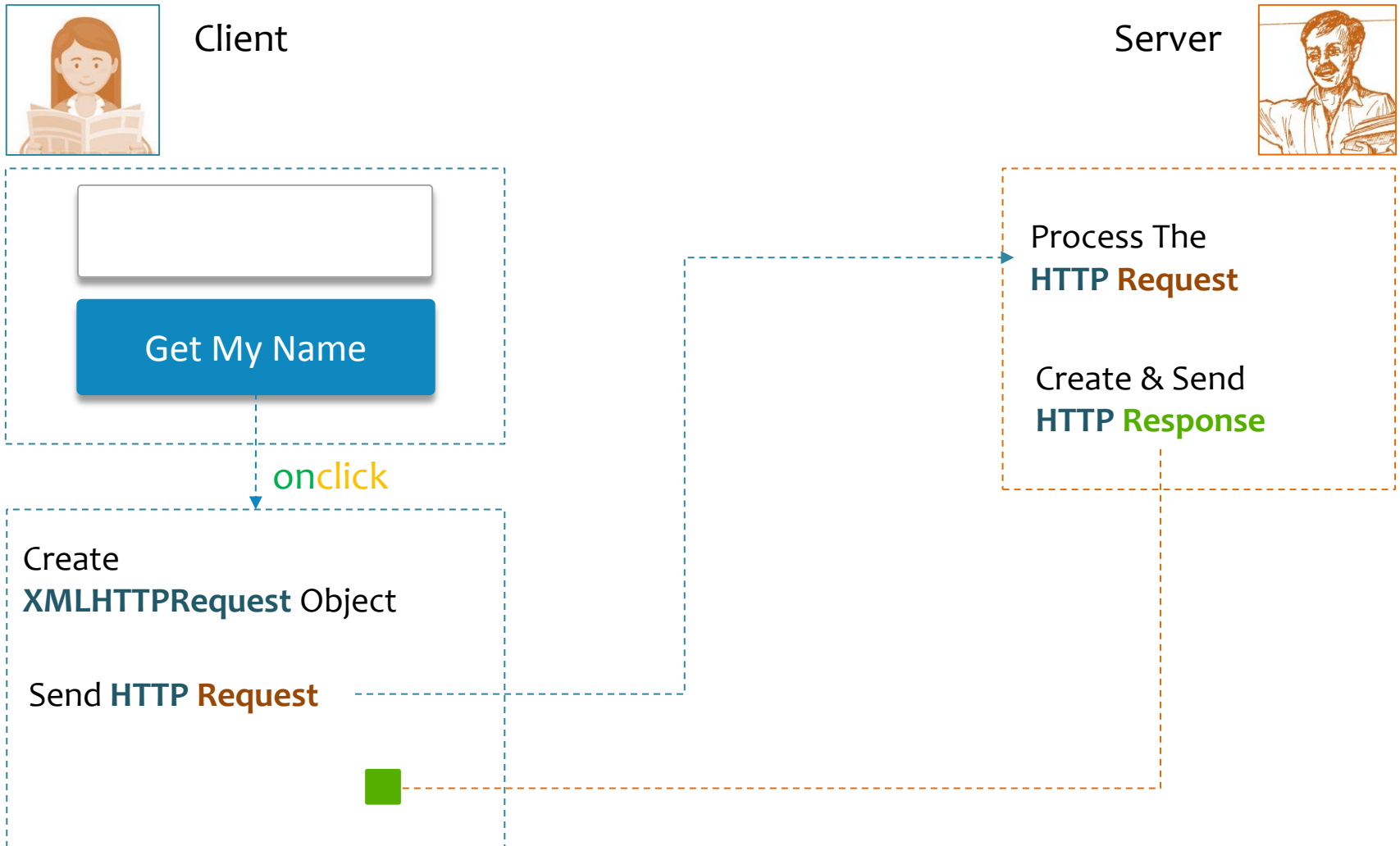
Server



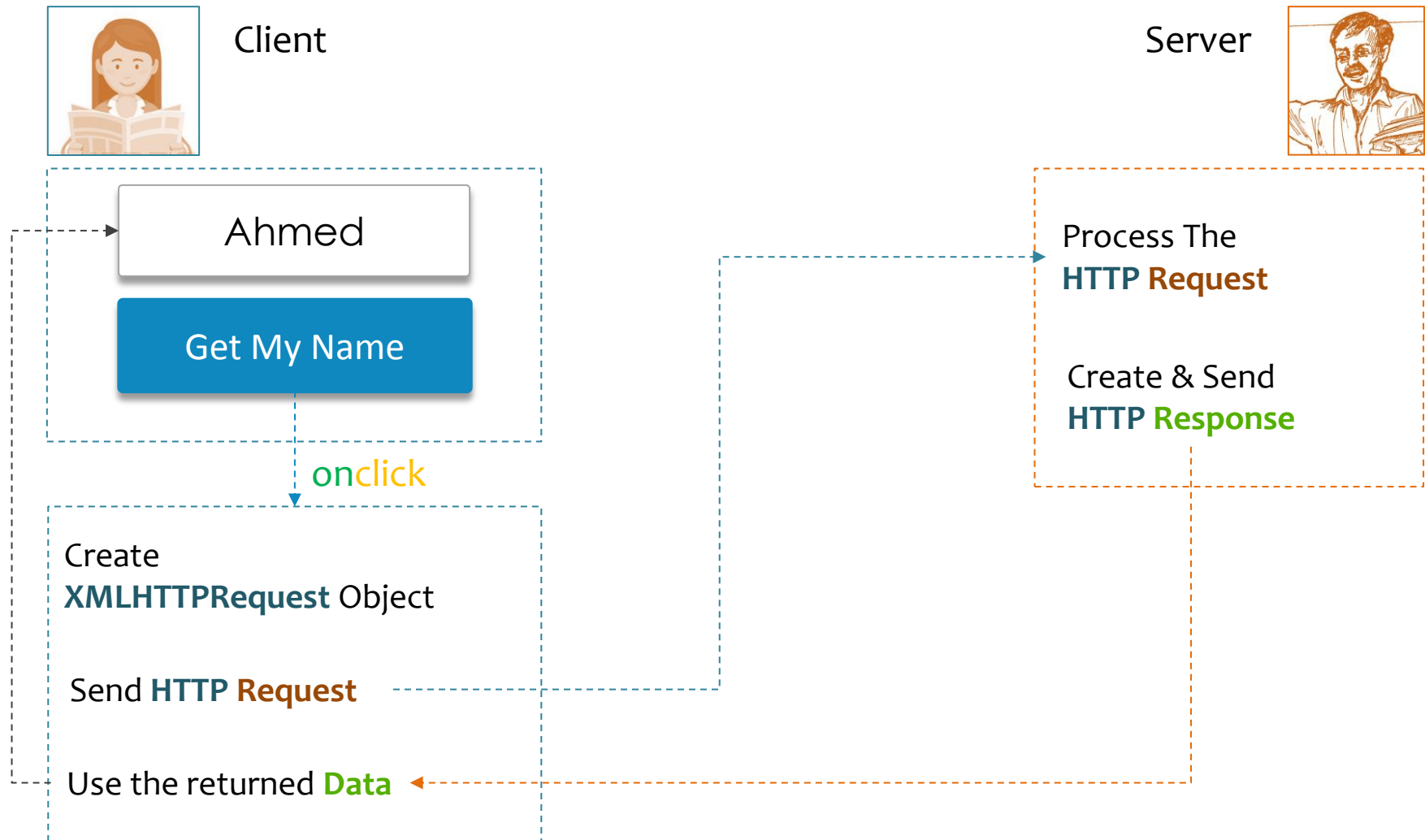
How does it work ?



How does it work ?



How does it work ?



JavaScript Implementation

Firstly we create a new object of XMLHttpRequest ...

```
var xhr = new XMLHttpRequest();
```

then ...

```
xhr.open(method, url, isAsync);
```

After that we send the data like ...

```
xhr.send();
```

or ...

```
xhr.send(string);
```



JavaScript Implementation

And for watching the state we use ... `xhr.onreadystatechange`

But What the request available states ... `xhr.readyState`

It may be `0` (`unsent`), `1` (`Opened`), `2` (`Headers Received`), `3` (`Loading`) or `4` (`Done`)

And the request status ... `xhr.status`

may be `200th` (`successful`),
`300th` (`Redirected`),
`400th` (`Request Errors` , you need to modify your request),
`500th` (`Server Errors` , You have no hand on it as a client)

And the server response will be come in ... `xhr.responseText`



JavaScript Example



Client

```
var xhr = new XMLHttpRequest();

xhr.open("get", "api.openweather.com/?country=FR", true);

xhr.send();

xhr.onreadystatechange = function() {

    if(this.readyState == 4 && this.status == 200) {

        console.log(this.responseText)

    }

}
```



jQuery Implementation

Again, what is jQuery Philosophy ...

Write less, Do more

```
$ .ajax ( ajaxObj )
```

Ajax Object contains the following properties and methods....

<code>url</code>	The url that the request redirect to.
<code>method</code>	The method name that the request will send by.
<code>data</code>	The data that sent with the request.
<code>success</code>	The callback function that will be executed when the response done.
<code>error</code>	The callback function that will be executed when there is an error
<code>complete</code>	The callback function that will be executed always.



jQuery Example



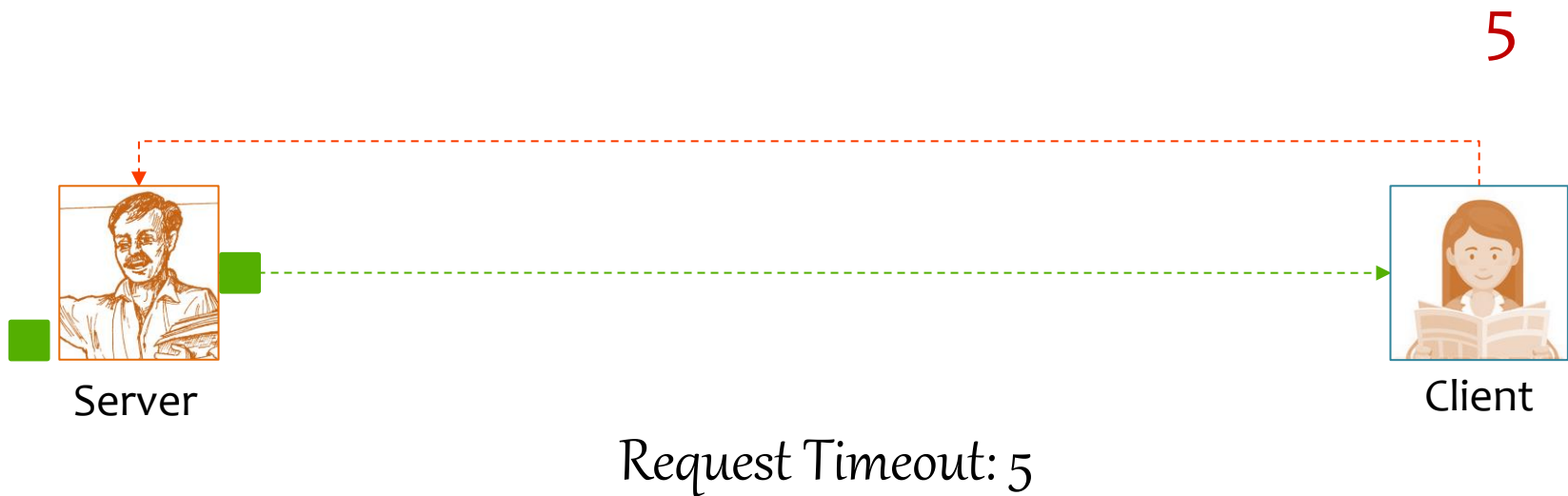
Client

```
$.ajax({  
    method: 'get',  
    async: true,  
    url: 'http://api.openweather.com',  
    data: {country: 'FR'},  
    success: function(res) {  
        console.log(res);  
    }  
})
```



Polling

Polling is technique to get the updated data from server once updated without reloading the page.





Client

```
function doPolling() {  
    $.ajax({  
        method: 'get',  
        async: true,  
        url: 'http://api.openweather.com',  
        data: {country: 'FR'},  
        success: function(res) {  
            console.log(res);  
            setTimeout(doPolling, 5000);  
        }  
    })  
}
```

doPolling()



Long Polling

Long Polling is an enhanced Polling technique to solve the problems of the basic polling method.





Client

```
function doLongPolling() {  
    $.ajax({  
        method: 'get',  
        url: 'http://localhost/myapi',  
  
        success: function(res) {  
  
            doLongPolling()  
        }  
    })  
}
```

doLongPolling()





Client

```
var lastModified = 0;

function doLongPolling() {

    $.ajax({

        method: 'get',

        url: 'http://localhost/myapi',

        data: {lastmod: lastModified},

        success: function(res) {

            lastModified = res.lastmod? res.lastmod:0;

            doLongPolling()

        }

    })

}

doLongPolling()
```



PHP Server



```
<?php
$file_name = 'document.txt';
$client_time = isset($_POST['lastmod']) ? $_POST['lastmod'] : 0;
$server_time = filemtime($file_name);
while($client_time >= $server_time){
    sleep(1);
    clearstatcache();
    $server_time = filemtime($file_name);
}
$response = array();
$response['lastmod'] = $server_time;
$response['msg'] = file_get_contents($file_name);
echo json_encode($response);
?>
```





```
import os,time

def get(self):

    clm = self.get_query_parameter('lastmod')
    slm, data = os.path.getmtime('file.txt'), {}

    while slm <= clm:

        time.sleep(3)

        slm = os.path.getmtime('file.txt')

    fl = open("file.txt", 'r')
    data['body'] = fl.read()
    data['lastmod'] = slm;
    fl.close()

    self.write(data)
```



to be continued